Conformal Prediction for Statistically Reliable LLM-as-a-Judge in Recommender Systems

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Abstract

LLM-as-a-judge has emerged as a promising strategy for evaluating recommender systems. Yet, treating simulated ratings as real feedback can make unreliable evaluation appear reliable. This risk is especially consequential in sensitive domains such as nutrition and health. In this paper, we treat the LLM-as-a-judge paradigm as a calibrated relevance simulation problem: an LLM-based simulator predicts human preferences from user and item representations, and conformal prediction converts them into uncertainty intervals. To account for the nature of recommendation, we introduce stratified conformal calibration by user rating variance. Experiments with three LLMs and seven conformal prediction methods show that point estimates of preferences hide failures on low-rating items, while conformal interval width is driven by user-level heterogeneity.

Keywords

LLM-as-a-Judge, User simulation, Conformal prediction

1. Introduction

Evaluating recommender systems remains difficult because the available evaluation options are either efficient but biased, or reliable but costly. Offline metrics such as recall and hit rate are inexpensive and repeatable, yet they are computed on historical interactions and reflect only the items users happened to encounter. Online experiments and user studies provide stronger evidence, but they are slow, expensive, and difficult to run for every design choice. Large Language Models (LLMs) can act as flexible judges of generated outputs and simulators of human preferences [1, 2, 3, 4, 5, 6]. To this end, existing methods provide the LLM with user/item data and ask it to produce a preference that approximates the judgment a user with that profile would provide [7, 8].

Unfortunately, LLM judgments may appear confident while remaining biased, miscalibrated, or misaligned, making them unreliable when used as evaluation signals [9, 10, 11]. This motivates the need to move from point judgments to uncertainty-aware judgments. Conformal prediction offers a way to do so: it is post-hoc, distribution-free, and provides finite-sample coverage guarantees under exchangeability [12, 13, 14]. Recent work has shown that it can transform LLM evaluations into prediction intervals under different tasks (e.g., text summarization) [15, 16, 17], but recommendation is not a routine application of this idea; it changes the calibration problem itself. In recommendation, the target rating is a joint property of a user–item pair: the same item may be relevant for one user and irrelevant for another. This makes uncertainty depend on rating behavior rather than only on the item. As a result, conformal prediction may merge stable users, who express consistent preferences, with volatile users, whose ratings span diverse tastes. The open issue is to determine whether conformal prediction can provide reliable uncertainty estimates when LLMs are used to simulate user preferences.

We study this problem in recipe recommendation, where unreliable preference simulation can lead to inappropriate results [18, 19]. Specifically, we frame the LLM-as-a-judge task as a calibrated relevance simulation problem: given a user/item representation and a target recipe, the LLM predicts the rating

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that the user would assign on a 1–5 scale. The simulated rating is treated as a proxy judgment, calibrated against held-out real ratings, and converted via conformal prediction into an interval that quantifies simulation uncertainty. Our contribution is twofold: we introduce user-stratified conformal calibration for LLM-based preference simulation and empirically show that its uncertainty is primarily driven by user-level rather than item-level rating variability.

This extended abstract summarizes results from a short paper accepted at CIKM 2026 [20].

2. Methodology

Human Judgment Simulation Let $\mathcal{U}$ and $\mathcal{I}$ denote the sets of users and items. For each pair $(u, i) \in \mathcal{U} \times \mathcal{I}$, let $y_{ui} \in \mathcal{Y} = \{1, \dots, K\}$ be the ground-truth rating on an ordinal scale with K levels. Each user u is represented by preference evidence r_u, and each item i by metadata m_i.

Let f_θ denote the user simulator. Given a user-item pair (u, i), it produces the score vector:

$$z_{ui} = f_\theta(r_u, m_i) = \left[z_{ui}^{(1)}, z_{ui}^{(2)}, \dots, z_{ui}^{(K)}\right] \in \mathbb{R}^K \tag{1}$$

where $z_{ui}^{(k)}$ represents the support for rating k. The simulated rating is then $\hat{y}_{ui} = \arg\max_{k \in \mathcal{Y}} z_{ui}^{(k)}$.

In our implementation, f_θ is instantiated by an LLM and z_{ui} corresponds to the raw logits of the valid rating tokens at the rating-generation position [15]. We use these scores as input to conformal calibration to construct a calibrated prediction interval for the true rating y_{ui}.

Post-hoc Conformal Calibration The purpose of calibration is to quantify the reliability of simulated ratings with respect to real user ratings. We use split conformal prediction to calibrate the outputs of f_θ.

Nonconformity Scoring. We consider a calibration set of real user–item interactions, $\mathcal{D}_{\text{cal}} = \{(u, i, y_{ui})\}_{j=1}^n$, reserved exclusively for calibration. For each example, the simulator produces $f_\theta(r_u, m_i) = z_{ui}$, yielding the simulator-calibration outputs $\mathcal{O}_{\text{cal}} = \{(z_{ui}, y_{ui})\}_{j=1}^n$. A nonconformity function $s(\cdot)$ then measures the disagreement between the simulator output and the ground-truth rating, producing $\mathcal{R}_{\text{cal}} = \{r = s(z_{ui}, y_{ui})\}_{j=1}^n$. A low nonconformity score indicates high support for the true rating, whereas a high score indicates stronger support for distant ratings. Different conformal methods instantiate s differently while following this principle.

Nonconformity Cutoff Computation. Calibration is controlled by the miscoverage level $\alpha \in (0, 1)$, which specifies the tolerated probability that the interval does not contain the true user rating, corresponding to a target coverage of $1 - \alpha$. To achieve this target, we use the calibration examples to learn how large the simulator's past errors have been. We sort the calibration nonconformity scores $\{r_j\}_{j=1}^n$ and choose a cutoff that retains the desired fraction of examples. We define $k_\alpha = \lceil (n+1)(1-\alpha) \rceil$ and set the conformal threshold as $\hat{q}_\alpha = r_{(k_\alpha)}$. This standard split-conformal finite-sample correction yields a slightly conservative threshold supporting the coverage guarantee.

Non-Stratified Rating Interval Construction. For an unseen user–item pair (u, i), the simulator produces $z_{ui} = f_\theta(r_u, m_i)$. We evaluate each candidate rating $y \in \mathcal{Y}$ and retain those whose nonconformity score does not exceed $\hat{q}_\alpha$, producing $C_\alpha(z_{ui}) = \{y \in \mathcal{Y} : s(z_{ui}, y) \le \hat{q}_\alpha\}$ which represents the calibrated set of plausible ratings. For example, the point prediction may be $\hat{y}_{ui} = 4$, while calibration may retain ratings $3, 4, 5$. Under exchangeability between calibration and test examples, the conformal set satisfies the finite-sample coverage guarantee, defined as follows:

$$1 - \alpha \le \mathbb{P}(y_{ui} \in C_\alpha(z_{ui})) \le 1 - \alpha + \frac{1}{n+1} \tag{2}$$

This means that the true user rating falls inside the calibrated set with the desired probability, up to the finite-sample correction. Narrow intervals indicate that the simulator provides a more confident approximation. For methods producing continuous interval bounds, we map them to the discrete rating scale. Following [15], this adjustment preserves the conformal coverage guarantee.

User-Stratified Rating Interval Construction. In recommendation, calibration error may vary substantially across users. Stable raters may require narrow intervals, while users with highly variable

preferences may need wider ones. A single global threshold $\hat{q}_\alpha$ can therefore mix different uncertainty regimes and produce intervals that are poorly adapted to user-level preference consistency. To account for this heterogeneity, we assign each user to a stratum according to their empirical rating standard deviation $h : \mathcal{U} \to \{1, \dots, G\}$. For stratum g, let $\mathcal{D}_{\text{cal}}^{(g)} = \{(u, i, y_{ui}) \in \mathcal{D}_{\text{cal}} : h(u) = g\}$ be the calibration subset with size n_g, and let $r_{(1)}^{(g)} \le r_{(2)}^{(g)} \le \cdots \le r_{(n_g)}^{(g)}$ denote the corresponding sorted calibration scores. We define $k_\alpha^{(g)} = \left\lceil (n_g + 1)(1 - \alpha) \right\rceil$ and the stratum-specific conformal threshold $\hat{q}_\alpha^{(g)} = r_{(k_\alpha^{(g)})}^{(g)}$.

For an unseen pair (u, i) with $h(u) = g$, the stratified interval is computed as:

$$C_\alpha^{(g)}(z_{ui}) = \left\{ y \in \mathcal{Y} : s(z_{ui}, y) \le \hat{q}_\alpha^{(g)} \right\} \tag{3}$$

Under exchangeability within each stratum, the coverage guarantee in Equation 2 holds separately for each group, with n_g calibration examples. This makes calibration sensitive to user heterogeneity and avoids forcing users with different rating behavior into a single global uncertainty model.

3. Experimental Evaluation

Our experiments are guided by the following research questions: (i) how reliable are LLM-simulated ratings as point estimates of user preferences in recommendation? (RQ1); (ii) which conformal prediction methods provide the best reliability–usefulness trade-off for LLM-simulated ratings? (RQ2); and (iii) how does user preference heterogeneity affect conformal uncertainty in recommendation? (RQ3).

Experimental Setup. We conduct experiments on HUMMUS [19], a recipe recommendation dataset containing interactions between 39,315 users and 507,335 recipes. Each interaction includes an explicit 1–5 star rating. Since user rating behavior in HUMMUS is heavily skewed toward high ratings, we adopt a stratified sampling strategy: we partition users into four subgroups based on their personal rating standard deviation ($[0, 0.3)$, $[0.3, 0.5)$, $[0.5, 0.7)$, and $[0.7, \infty)$), and sample 1,000 users per subgroup, resulting in 4,000 users in total. For each user, we use the two most recent interactions for testing, the previous two for calibration, and the preceding 20 as prompt history.

We evaluate three open-source LLMs: Qwen2.5-32B-Instruct [21], LLaMA-3.1-8B-Instruct [22], and Mistral-7B-Instruct-v0.3 [23], spanning a range of parameter counts (7B–32B) and training methods. The prompt includes the preceding 20 rated interactions and their recipe features.

We adopted the same suite of methods designed to construct prediction intervals [15]. Specifically, we include conformalized quantile regression (CQR) [16] and its asymmetric variant [24], CHR [25] which employs a histogram-based quantile estimator, LVD [26] which relies on kernel regression, two boosted conformal prediction methods [27] (Boosted CQR and Boosted LCP), and R2CCP [17]. All methods are applied with $\alpha = 0.10$, targeting 90% empirical coverage. Results are averaged over 30 random seeds to account for the stochasticity of methods involving internal model fitting.

Experimental Results.[1] We first compute the point-prediction performance of the three LLM judges **(RQ1)**. At the aggregate level, the models appear comparable: accuracy ranges from 0.709 to 0.754, and MAE ranges from 0.340 to 0.382. However, this aggregate view hides a severe imbalance across rating levels. Per-rating MAE shows that all models perform poorly on low-rating recipes, with errors above 3.6 when the true rating is 1. On a 1–5 scale, this indicates that the judges often assign very high scores to recipes that users actually rated very poorly. In contrast, MAE drops to 0.15–0.22 for recipes with true rating 5. These results show that aggregate performance is driven by the majority high-rating class and can overstate the reliability of LLM-simulated ratings.

The failures observed in the first analysis motivate moving beyond single-rating predictions and quantifying uncertainty through conformal intervals **(RQ2)**. Most methods reach the 90% coverage target, with Boosted CQR being the only method that consistently falls below it across all judges (88.06–89.04%). However, achieving coverage is not sufficient: CQR variants and CHR systematically over-cover (~95–96%), which indicates conservative and unnecessarily wide intervals. Among the

[1]The complete analysis is available at https://github.com/maddalena-amendola/conformal_prediction_recsys.

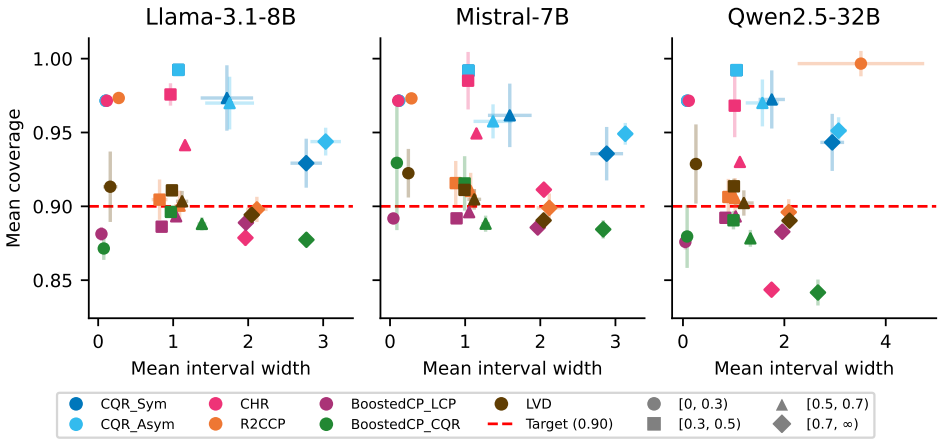


Figure 1: Coverage–efficiency trade-off across user-variance groups.

methods that satisfy the coverage target, R2CCP provides the best efficiency, reaching approximately 90% coverage with the narrowest intervals (width < 1) across all judges. To understand what drives uncertainty under global calibration, we analyze the association between R2CCP interval width and two sources of rating variability: recipe-level variance and user-level variance. We find that (i) recipe rating variance has a negligible practical association with interval width, despite statistical significance ($\rho \leq 0.06$, $p < 0.001$), and (ii) there is a strong and consistent positive association between user rating variance and interval width ($\rho = 0.48$–0.62, $p < 0.001$). Users with more variable historical ratings receive substantially wider intervals than consistent raters. This indicates that conformal uncertainty in recommendation is driven by user preference heterogeneity rather than recipe-level variability, motivating the user-stratified analysis.

Finally, to answer **RQ3**, we evaluate conformal prediction within the four user groups defined by historical rating standard deviation: $[0, 0.3)$, $[0.3, 0.5)$, $[0.5, 0.7)$, and $[0.7, \infty)$. Each group contains 1,000 users, corresponding to 2,000 calibration pairs and 2,000 test pairs. A two-sample Kolmogorov–Smirnov test detects no significant calibration–test shift within any group ($p > 0.05$). Figure 1 reports the coverage–efficiency trade-off for all methods across user groups and LLM judges. As user preference variance increases, intervals become substantially wider: widths range from approximately 0–0.5 for the lowest-variance group to nearly 3.0 for the highest-variance group. On a 1–5 scale, such wide intervals cover most of the rating range and are weakly actionable. R2CCP and Boosted LCP remain closest to the 90% coverage target across groups and judges, while CQR variants and CHR tend to over-cover, especially for low-variance users, producing unnecessarily conservative intervals. Boosted CQR is the least reliable, often falling below the target in higher-variance groups. Overall, no method is optimal across all groups: as preference heterogeneity increases, methods either lose coverage or produce excessively wide intervals. R2CCP offers the most stable trade-off, especially for low- and mid-variance users, confirming the value of the user-stratified calibration.

4. Conclusions and Future Works

In this paper, we studied conformal prediction for LLM-as-a-judge in recommendation, showing that valid coverage is achievable but interval usefulness depends strongly on user preference heterogeneity. Intervals are informative for consistent users, while they become broad and weakly actionable for users with highly variable tastes. This supports user-stratified calibration as a principled strategy for uncertainty-aware recommendation. Future work should test this behavior in sensitive domains beyond food, develop adaptive calibration strategies that account for user heterogeneity more finely, and study how calibrated intervals can guide evaluation pipelines.

GenAI Usage Disclosure

Generative AI tools were used to assist with language editing and limited drafting during the preparation of this paper. All scientific content, analyses, results, and final text were critically reviewed and validated by the authors.

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